

Authors' Response to Reviews of

Neuromorphic Computing Based on Parametrically-Driven Oscillators and Frequency Combs

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RC: Reviewers' Comment, AR: Authors' Response,

Dear Reviewers,

Thank you for the reviews. The comments led us to add several diagnostics that we agree the original submission lacked, and these have materially sharpened the mechanistic account of why the parametric resonance regime outperforms the comb regimes. All required corrections have been made to the manuscript and the supplemental material. Additions to the main text are marked in red; the supplemental material has been extended with eight new sections.

Section numbers below refer to the revised supplemental material.

1. Reviewer #1

RC: *The manuscript adopts one-step-ahead prediction of chaotic time series as the benchmark for reservoir performance. While this is a widely used and reasonable choice, it conflates several underlying computational properties, such as short-term memory and nonlinear transformation capability, into a single aggregate metric (NMSE), making it difficult to disentangle which specific computational resource is responsible for the observed differences between regimes. I would recommend that the authors additionally evaluate the information processing capacity (IPC) proposed by Dambre et al. [Sci. Rep. 2, 514 (2012)]. By decomposing reservoir capacity into contributions from different nonlinear degrees and delays, this approach would allow the authors to identify precisely which computational capabilities are enhanced or lacking in the parametric resonance regime compared to the frequency-comb regime.*

AR: We agree, and we have carried out full IPC decomposition. The results are in Sec. II of the supplemental material and are summarized in the second paragraph of page 3 of the main text.

The decomposition requires an i.i.d input, since Legendre basis is orthogonal only under that measure and driving with Mackey-Glass would violate the orthogonality the method depends on. Therefore, we used $u(n) \sim \mathcal{U}[-1, 1]$, i.i.d., with the drive built from $(u+1)/2$ and the targets taken from u directly. The basis caps total degrees at 11, delays at 25, and at most three distinct delays per target. This gives 447,161 basis functions. The estimator was validated against a 20-node linear delay line, where it returns a total of 20.175 against a measured rank of 20 (0.87% relative error), with 20,000 recovered at degree one.

	Parametric resonance	Coherent comb	Chaotic comb
Total capacity	160.32	18.0 (16.9–18.2)	5.54 (5.49–5.81)
Fraction of the bound $r = 256$	62.6%	7.0%	2.2%
Capacity per available dimension	0.626	0.070	0.022
Ratio to parametric resonance	–	$8.9\times$ lower	$29\times$ lower

Table 1: Comparison of nonlinear memory capacity across the three regimes.

The regimes differ in total capacity as well as where that capacity sits. At parametric resonance the per-degree contribution rises across the whole basis: 2.07, 3.29, 4.88, 6.92, 9.28, 12.21, 15.60, 19.35, 23.75, 28.82, 34.13. It is still rising at degree 11, the highest degree scanned. The two comb regimes behave differently from each other. The coherent comb neither deepens nor vanishes, staying flat between 1.5 and 1.8 at every degree, while the chaotic comb decays steadily and falls to within a factor of two of the false-positive expectation beyond degree three.

Both comb regimes are therefore shallower than parametric resonance, and both lose the high-degree nonlinear capability that one-step prediction of a chaotic series requires. As any finite degree cap truncates the sum, these totals are lower bounds and the measurement establishes the ordering and the accessible depth rather than a saturated total.

RC: *To better understand why the frequency-comb regime underperforms relative to naive expectations, I would suggest that the authors examine two additional diagnostic quantities: the echo-state property (ESP) and the effective dimensionality of the reservoir state. The ESP can be quantified by measuring how quickly trajectories of the reservoir, initialized from different initial conditions but driven by the same input sequence, converge onto a common trajectory; faster convergence indicates a strong echo-state property. The effective dimensionality, on the other hand, characterizes how many independent degrees of freedom are actually being used within the 2048-dimensional reservoir state and can be measured by principal component analysis. In general, good computational performance requires both a strong echo-state property and sufficiently high dimensionality, but these two quantities are known to trade off against each other, so an appropriate balance between them is essential. I expect the frequency-comb regime to exhibit higher effective dimensionality than the parametric resonance regime. However, given its inferior computational performance, I would hypothesize that this comes at the cost of a degraded echo-state property. I recommend that the authors directly measure both quantities across the sub-threshold, parametric resonance, and comb regimes, as this would provide a clear mechanistic account of the observed performance differences.*

AR: This hypothesis is what we find. Both quantities are now measured across all four regimes in Sec III of the supplemental material.

Effective dimensionality is quantified by the participation ratio $D_{\text{eff}} = (\sum \lambda)^2 / \sum \lambda^2$ over the covariance eigenvalues of the standardized feature matrix. This requires no arbitrary variance cutoff and satisfies $1 \leq D_{\text{eff}} \leq r$. The echo state property is measured by driving 12 replicas from different initial state conditions with an identical input sequence and fitting $\ln d(n) = \ln d(0) - \lambda n$ to the mean pairwise separation. The distance accounts for the Z_2 symmetry $(\psi_1, \psi_2) \rightarrow (\psi_1, -\psi_2)$ in Eqs. (1) and (2) in the main text. Without this, parametric resonance could appear to violate the echo-state property because replicas may settle on opposite branches. These branches are equivalent and cannot be distinguished by the Z_2 -invariant readout.

Table 2: Effective dimensionality, variance concentration, active features, and convergence across the four dynamical regimes.

	Sub-threshold	Parametric resonance	Coherent comb	Chaotic comb
D_{eff}	1.915	2.770	7.147	7.403
Components for 95% / 99% variance	2 / 2	8 / 11	30 / 89	54 / 179
Leading component, variance fraction	0.607	0.565	0.310	0.307
Active features (of 2048)	1024	2048	2048	2048
Convergence rate λ (per symbol)	4.99	4.03	2.5×10^{-4}	2.1×10^{-4}
Exponential fit R^2	1.000	0.974	0.011	0.004

Effective dimensionality rises monotonically into the comb regimes, $1.915 \rightarrow 2.770 \rightarrow 7.147 \rightarrow 7.403$. On the other hand, the echo-state property collapses. Parametric resonance converges rapidly, with a rate of 4.03 per symbol and an exponential fit of $R^2 = 0.974$. Neither comb regime shows clear convergence. The fitted rates of 2.5×10^{-4} and 2.1×10^{-4} per symbol have very low R^2 values of 0.011 and 0.004, respectively. Therefore, the exponential model does not describe the comb regimes well, and these rates should not be interpreted as slow convergence.

Below threshold, $\psi_1 \equiv -$, so the 1024 mode-2 columns are constant and dropped. This leaves 1024 active features. The low D_{eff} is not a consequence of having half as many active features. The constant mode-2 columns carry no variance and therefore do not contribute to the participation ratio. Instead, the low D_{eff} reflects the low dimensionality of the surviving mode-1 features: the first two components account for 60.7% and 39.3% of the variance, respectively, capturing 99.94% in total and providing $D_{\text{eff}} = 1.91$.

Between parametric resonance and the coherent comb, D_{eff} increases by $2.5\times$, while the number of components needed to capture 99% of the variance increases by $8\times$. This difference shows that much of the additional dimensionality in the comb lies in weakly excited directions. These directions contribute little to the participation ratio and are strongly suppressed by the readout regularization. Thus, the combs provide richer spectra and higher dimensionality, but with poorer state convergence, which helps explain their lower prediction performance.

RC: *In Fig. 2(a) (and also in the caption of Fig. 1), the modulation depth is stated as $\Delta F = 0$. However, according to Eq. (3), $f(n) = F_{\text{avg}} + \Delta F \times (s(n) - 0.5)$, setting $\Delta F = 0$ renders the drive amplitude $f(n)$ completely independent of the input symbol $s(n)$. Under this condition, no information about the input time series would be transmitted into the reservoir via the drive amplitude, which would make the subsequent prediction task meaningless. Since the manuscript nonetheless reports successful prediction, I suspect this is a typographical error.*

AR: The reviewer is correct that the label was wrong. We thank them for pointing this out. It was a mislabelling, not a typographical error. The input was always included in the drive.

Throughout the study, we use the input-proportional encoding

$$f(n) = F_{\text{avg}}(s(n) + 0.5), \quad (1)$$

so the drive ranges from $0.5F_{\text{avg}}$ to $1.5F_{\text{avg}}$. This is equivalent to Eq. (3) with the modulation depth set to $\Delta F = F_{\text{avg}}$:

$$F_{\text{avg}}(s(n) + 0.5) = F_{\text{avg}} + F_{\text{avg}}(s(n) - 0.5). \quad (2)$$

Thus, reporting $\Delta F = 0$ was inconsistent with the definition of ΔF as the modulation depth.

We have corrected this throughout the manuscript. All occurrences of “ $\Delta F = 0$ ” have been changed to “ $\Delta F = F_{\text{avg}}$ ”. This includes the caption of Fig. 1, the three panels of Fig. 2, and the parameter notes for Figs. 3–5. The methodology now also states that $\Delta F = F_{\text{avg}}$ is the default encoding unless otherwise specified.

This correction also required us to revise the discussion of Fig. 2(a). Since $\Delta F = F_{\text{avg}}$ varies with F_{avg} and reaches 50 along the horizontal axis, panel (a) does not represent the smallest modulation depth. Instead, its modulation depth increases with F_{avg} . We have therefore removed the statement that ΔF increases from zero and clarified that using a fixed ΔF removes this dependence on F_{avg} , producing a more uniform low-NMSE region.

RC: *In constructing the reservoir feature vector, the authors include the discrete Fourier transform (DFT) of the sampled ψ_1 and ψ_2 trajectories, in addition to their time-domain trajectories. It is not entirely clear from the text whether this DFT is computed over the short trajectory sampled within a single symbol period of length T_{symbol} , or over the full time series spanning multiple symbol periods. I would ask the authors to explicitly clarify the exact time window over which this DFT is computed.*

AR: The transform is computed over a **single symbol period**, not the full record. We have clarified this in the methodology and in Sec. IV of the supplemental material.

For each symbol n and mode $j = 1, 2$, the transform uses the same $N_v = 512$ virtual-node samples as the time-domain features,

$$\tau_k = \frac{kT_{\text{symbol}}}{N_v}, \quad (3)$$

with $T_{\text{symbol}} = 5.0$ in normalized time. The frequency-bin spacing is therefore $1/T_{\text{symbol}} = 0.2\tau^{-1}$. Thus, the transform does not capture timescales longer than one symbol; any memory across symbols is carried by the state ψ .

The transform is applied to the **complex modal amplitudes** ψ_1 and ψ_2 , not to their sampled powers. The spectral features are

$$|\mathcal{F}[\psi_j]|, \quad (4)$$

with the magnitude taken after the transform. This preserves intra-symbol phase information that is lost in $|\psi|^2$. Since ψ_j is complex, its spectrum is two-sided and asymmetric, so the positive and negative frequency bins can contain different information. With $N_{\text{FFT}} = N_v$, the full two-sided spectrum is retained. If the transform were instead applied to the sampled power, the signal would be real and its spectrum would be conjugate-symmetric, making the spectral features largely redundant with the time-domain features.

RC: *The governing equations (1) and (2) are presented without an explicit derivation. Given that these equations form the theoretical foundation of the entire study, I would recommend that the authors include a detailed, self-contained derivation in the supplemental material.*

AR: A complete, self-contained derivation of Eqs. (1)–(2) in the main text has been added to Sec. I of the supplementary material.

RC: *The manuscript does not explicitly state whether, during the testing phase, the predicted output is fed back into the drive amplitude encoding to autonomously generate the subsequent time steps in a closed-loop (free-running) fashion, or whether the true time series is used to drive the reservoir at every time step even during testing (open-loop prediction). This distinction is important for interpreting the reported NMSE values: closed-loop generation would be expected to accumulate error over time, whereas open-loop evaluation measures only the local, single-step predictive accuracy of the readout at each time point. I would ask the authors to explicitly clarify which protocol was used.*

AR: The prediction is performed in an **open-loop (teacher-forced)** setting. The true time series is used to drive the reservoir during both training and testing, and the predicted output is never fed back into the drive. Thus, the reported NMSE measures single-step prediction error and does not include error accumulation from free-running prediction. We have clarified this in the methodology.

2. Reviewer #2

RC: *Redundancy in feature extraction: The feature vector combines time-domain samples (1024 dimensions) and Fourier magnitudes (1024 dimensions) from the same symbol period trajectories. Unless the sampling rate violates Nyquist, the spectral features introduce little independent information. The authors should demonstrate their added value via ablation studies or rank analysis; otherwise, the high dimensionality (2048) risks overfitting despite ridge regularization.*

AR: We have performed both analyses and report the results in Sec. IV of the supplementary material.

For the ablation study, all readouts use the same reservoir realisation, trajectory, warm-up period, and train/test split. Only the input feature block is changed. Each case uses its own cross-validated regularization parameter. The results are summarized in Table III of the supplementary material.

At parametric resonance, combining the time and spectral features gives a $7.35\times$ lower NMSE than the time features alone and a $4.5\times$ lower NMSE than the spectral features alone. This shows that the two feature blocks provide complementary information.

We also computed the principal angles between the leading 20 principal directions of the two standardized feature blocks. At parametric resonance, the canonical correlations range from 1.000 to 0.001, with a median of 0.641, showing that the two feature spaces overlap but are not identical. The full results are given in Fig. 4 of the supplementary material.

The sub-threshold case provides an important contrast. Its median canonical correlation is only 0.040, but combining the two blocks gives almost no improvement ($1.02\times$), because $\psi_2 \equiv 0$ makes half of the features inactive. Thus, the two analyses together show that the spectral features provide useful additional information in the operating regime.

The effective degrees of freedom and the effect of regularization are discussed in Sec. VII of the supplementary material.

RC: *No quantitative criterion for distinguishing coherent vs. chaotic combs: The boundary between these regimes is not defined. A metric (e.g., phase diffusion coefficient, spectral line width, or Lyapunov exponent) must be specified, and the exact parameter values for the chaotic-comb examples in Fig. 1(k-o) must be provided in the caption.*

AR: We have adopted the **largest Lyapunov exponent** as the primary measure for distinguishing coherent and chaotic combs. The exponent is computed at constant drive using the algorithm of Benettin et al., as described and validated in Sec. V of the supplementary material.

Two trajectories with an initial separation of $d_0 = 10^{-7}$ are evolved, with the separation rescaled to d_0 after each interval. This value is well above the integration tolerance of 10^{-10} and sufficiently small for the linearized dynamics to remain valid.

The classification uses $\lambda_{\max} > 10^{-2}$ for chaos and $|\psi_2| < 10^{-6}$ for the sub-threshold regime. The threshold is well above the $\sim 10^{-3}$ scatter of finite-time estimates for a true zero exponent. The method is also validated by the sub-threshold case, where the measured exponent recovers the known value $-\gamma_1 = -1$ to four decimal places. As an independent check, the normalized spectral entropy of the mode-2 power trace gives the same ordering of the three excited regimes.

We also mapped $\lambda_{\max}$ over the full $(F_{\text{avg}}, \Delta_1)$ parameter plane. The results are given in Sec. V of the supplementary material. Overall, 36.14% of the grid is sub-threshold, 41.35% is coherent, and 22.51% exceeds the chaos threshold. The chaotic-comb region, where both the comb and chaos conditions are satisfied, covers 16.21% of the grid and forms a connected region with $\lambda_{\max}$ ranging from 0.011 to 1.731.

The parameters for the operating points in Fig. 1 in the main text are now given in its caption. For the chaotic comb in Fig. 1(k-o), $F_{\text{avg}} = 20.0$, $\Delta_1 = +2.0$, and $\Delta_2 = -8.0$, with $\kappa = -9.0$ and $\gamma_{21} = 1.0$. The coherent comb uses $F_{\text{avg}} = 40.0$, $\Delta_1 = +4.5$, while the parametric resonance point uses $F_{\text{avg}} = 40.0$, $\Delta_1 = -4.5$, with the same κ and γ_{21} .

RC: *Data rate not rescaled to the intrinsic timescales of benchmark systems: The Mackey-Glass, Rössler, and Lorenz*

systems have widely different characteristic times (nearly an order of magnitude), yet all are fed at the same rate of 2424 samples/s. This leads to vastly different effective memory capacities across tasks. The authors should either rescale the symbol duration for each system or explicitly justify the single rate and discuss its impact.

AR: The rescaling requested by the reviewer was already applied, but this was not clear from our description. We have clarified this in the revised manuscript.

The three benchmarks are subsampled before being used as inputs, giving 17.0, 7.5, and 12.0 samples per characteristic time for Mackey–Glass, Lorenz, and Rössler, respectively. Thus, the difference in effective sampling rates is $2.3\times$, rather than the much larger difference between their raw timescales. The common drive rate therefore follows from this pre-normalization.

We also evaluated the effect of the data rate in the supplementary material using sweeps over the $(F_{\text{avg}}, \Delta_1)$ plane at low, intermediate, and high rates. The results show that performance depends on the match between the input timescale and the reservoir dynamics. At low rates, the dynamics settle between inputs and memory is lost. At high rates, the system does not respond fully within one symbol. The best performance occurs at intermediate rates, where the dynamics from successive inputs overlap. This behavior is consistent across all three benchmarks and for both ridge and LASSO readouts.

RC: *Initiation protocol in the bistable region not quantified: The authors note that performance in the bistable region depends on initialization, but they do not report how often the 50-period constant-amplitude drive actually selects the parametric branch. A sensitivity analysis to initial conditions and a more robust startup method (e.g., chirped drive) should be considered.*

AR: We have quantified the sensitivity to the initial conditions and adopted the more robust startup protocol suggested by the reviewer. The analysis is given in Sec. VI of the supplementary material.

At $\Delta_1 = -4.5$, the Arnold-tongue threshold and the upper stability threshold are $f_{\text{Arnold}} = 7.875$ and $f_{\text{upper}} = 26.03$, respectively. Taking the modulation of the drive into account gives a bistable range of

$$15.75 \leq F_{\text{avg}} \leq 17.35. \quad (5)$$

We tested 32 initial conditions at four amplitude scales, $\sigma \in \{0.01, 0.1, 1, 8\}$, over 24 values of F_{avg} and compared constant and ramped startup protocols. Within the bistable range, at the default scale $\sigma = 0.01$, the constant protocol selected the parametric branch in 0% of trials, whereas the ramped protocol selected it in 100% of trials. The F_{avg} values giving 50% selection were 23.4, 20.7, 17.3, and 11.5 for $\sigma = 0.01, 0.1, 1$, and 8, respectively. Above $F_{\text{avg}} \approx 24.6$, both protocols agree because the trivial branch is no longer stable.

Within the bistable range, the constant protocol therefore leaves the reservoir on the trivial branch, where $\psi_2 \equiv 0$ and half of the features are inactive. We have consequently adopted the ramped protocol, which starts above f_{upper} and then decreases adiabatically to the target F_{avg} .

At the parametric-resonance point, $F_{\text{avg}} = 40$ and the peak drive is 60, well above $f_{\text{upper}} = 26.03$. Both protocols then converge to $|\psi_2| = 8.002$, with agreement to five significant figures across 32 initial conditions and all four amplitude scales.

We also corrected the startup procedure by seeding the random-number generator and repeated the reported calculations over 20 realizations. The resulting variation is small, with at most 5.6% variation in total capacity and 0.96% in effective dimensionality. Parametric resonance is identical across realizations, while interquartile ranges are reported for the comb regimes.

RC: *Insufficient details on regularization and numerical integration: The ridge penalty $\lambda = 10^{-3}$ is fixed without cross-validation, despite the feature dimension (2048) greatly exceeding the training sample size (800). The choice of λ critically affects generalization. Also, the adaptive Runge-Kutta integrator is not verified for stiff regimes (e.g., chaotic combs); the error tolerance and steps per symbol period should be reported.*

AR: Both points are well taken. The regularization and numerical integration details are given in Secs. VII and VIII of the supplementary material.

Regularization. We now select λ using five-fold cross-validation with contiguous, non-shuffled folds. This avoids leakage between nearby time-series samples. The search covers 10^{-8} to 10^6 , with further extension if the optimum lies at a boundary, followed by local refinement.

The effective degrees of freedom,

$$\nu(\lambda) = \text{tr}(H) = \sum_i \frac{s_i^2}{s_i^2 + \lambda}, \quad (6)$$

show that the effect of the fixed penalty depends strongly on the regime. At $\lambda = 10^{-3}$, ν is 63.7 at parametric resonance but about 700 in the two comb regimes, despite having only 750 training samples. Cross-validation reduces ν to 30.4, 225.5, and 164.5 for parametric resonance, coherent comb, and chaotic comb, respectively. The corresponding test NMSEs improve from 1.33×10^{-3} to 4.51×10^{-4} , from 6.67×10^{-2} to 1.28×10^{-2} , and from 7.16 to 6.36×10^{-2} . The full results are given in Sec. VII of the supplementary material.

Importantly, the main result is unchanged. At their respective optima, the coherent comb still has a $28\times$ higher NMSE than parametric resonance. Across the full parameter sweep with per-point cross-validation, parametric resonance also outperforms the coherent comb by $8.0\times$. We therefore use cross-validation for the reported results.

Numerical integration. Eqs. (1)–(2) in the main text are integrated as a real four-component system using an adaptive fifth-order Runge–Kutta method with relative tolerance 10^{-8} and absolute tolerance 10^{-10} . The terminal state is carried forward between symbols. Since the step size is adaptive, we report the number of right-hand-side evaluations per symbol in Sec. VIII of the supplementary material.

We also compared the explicit method with DOP853, Radau, and BDF at the same tolerances. The implicit methods were between $2.28\times$ and $3.93\times$ more expensive, including for the chaotic comb. These results indicate that the system is not stiff in the regimes studied and that the adaptive explicit method is appropriate.

We thank both reviewers again. The diagnostics requested here have converted what was previously an empirical observation that the comb regimes underperform despite richer spectra into a mechanistic account: the combs gain dimensionality, but in weakly excited directions, and at the cost of the state convergence that reservoir computation requires.

Sincerely,
Mahadev Sunil Kumar and Adarsh Ganesan